Percussion Frequencies

The following document outlines two **Markdown tables** with reference frequencies for:

- **Standard drumkit (5-piece)**
- **Multiple percussion instruments**

These references can be used to inform in testing the model

Glossary

Fundamental/Body: This refers to the core, lowest, and often most prominent pitch or "weight" of an instrument's sound. It's what gives a drum its "thud" or a maraca its central "shhh." Boosting this area adds fullness and perceived power.

Muddiness/Boxiness: These terms describe undesirable resonant frequencies, often in the low-mids to mids (around 150 Hz to 800 Hz), that can make an instrument sound unclear, "honky," or like it's trapped in a box. Cutting these frequencies can clean up the sound.

Beater Attack / Attack / Snap / Slap / Click / Crack: These all refer to the initial transient impact of a sound. It's the sharp "click" of a kick drum beater, the "snap" of a snare, or the "slap" of a conga. These high-frequency components (often 2 kHz and above) provide definition and cut through a mix.

Ringing: Specifically for drums (like snare or toms), "ringing" refers to sustained, often undesirable, resonant frequencies that linger after the initial hit. This can make a drum sound uncontrolled or out of tune.

Sizzle / Crispness / Sheen / Sparkle / Air / High Harmonics: These terms describe the very high-frequency content (typically from 5 kHz all the way up to 20 kHz and beyond) that gives instruments their brilliance, clarity, and sense of space. "Sizzle" often refers to hi-hats or cymbals, while "sheen" and "sparkle" are common for cymbals and other bright percussion. "Air" is the very top end that adds openness and realism.

Body/Warmth/Depth/Fatness:
These terms refer to the fundamental frequencies and the lower-mid range (roughly 80 Hz to 500 Hz) that give an instrument its foundational richness, fullness, and perceived power. Boosting here can make a sound feel more substantial.

Main Tone / Clang / Click / Scratch:
These describe the dominant, characteristic sound or timbre of specific percussion instruments. For example, the "clang" of a cowbell, the distinct "click" of claves, or the "scratch" of a guiro. These often reside in the mid-range to high-mid frequencies, giving the instrument its unique identity.

Standard Drumkit

Drum Part	Aspect	Typical Frequency Range (Hz)	Notes & Characteristics
Kick Drum			
	Fundamental/Body	40 - 150	*Thud* or *weight*, often 50-80 Hz for core impact
	Muddiness/Boxiness	100 - 400	Frequencies that can sound muddy or boxy; often targeted for cuts
	Beater Attack	2k - 5k	Presence and click of the beater hitting the head (can extend to 7k)
Snare Drum			
	Body/Fatness	150 - 250	The main thwack or fundamental tone, adds warmth and impact
	Ringing/Muddiness	250 - 1.5k	Unwanted ringing or boxy resonance
	Attack/Snap	3k - 5k	Crispness, initial impact, and *crack* of the stick
	Sizzle/Air	8k+	From the snare wires, adds brightness and air (often a high shelf boost)
Toms			
	Body/Resonance	60 - 200	Deeper for floor toms (e.g., 70-90 Hz), higher for rack toms (e.g., 120-200 Hz)
	Muddiness/Boxiness	250 - 1k	Resonances that might need cutting (e.g., 400 Hz for floor tom ring, 600 Hz for rack tom boxiness)
	Attack	1k - 5k	Stick attack and clarity for definition
Hi-Hats			
	Fundamental/Body	300 - 3k	The main *shhh* sound, often dominant around 2-3 kHz
	Crispness/Sizzle	8k - 17k+	Brightness, *air,* and metallic shimmer
Cymbals (Crash, Ride)			
	Body/Clang	300 - 600	Lower for large crashes, fundamental metallic sound or *gong* resonance
	Sheen/Sparkle	4k - 12k	For ride cymbal definition, crash cymbal brilliance
	Air/High Harmonics	10k - 20k+	Openness, shimmer, and extended high-frequency content

Percussion

Instrument Part	Aspect	Typical Frequency Range (Hz)	Notes & Characteristics
Kick Drum			
	Fundamental/Body	*40 - 150*	*Thud* or *weight,* often 50-80 Hz
	Muddiness/Boxiness	*100 - 400*	Frequencies to cut to remove unwanted resonances
	Beater Attack	*2k - 5k*	Presence and click of the beater (can extend to 7k)
Snare Drum			
	Body/Fatness	*150 - 250*	Main *thwack* or fundamental tone
	Ringing/Muddiness	*250 - 15k*	Frequencies to cut for unwanted ring or boxiness
	Attack/Snap	*3k - 5k*	Crispness and initial impact
	Sizzle/Air	*8k+*	From the snare wires (often high shelf)
Toms			
	Body/Resonance	*60 - 200*	Deeper for floor toms, higher for rack toms
	Muddiness/Boxiness	*250 - 1k*	Resonances that might need cutting
	Attack	*1k - 5k*	Stick attack and clarity
Hi-Hats			
	Fundamental/Body	*300 - 3k*	Main *shhh* sound, often dominant around 2-3 kHz
	Crispness/Sizzle	*8k - 17k+*	Brightness and air
Cymbals (Crash, Ride)			
	Body/Clang	*300 - 600*	Lower for large crashes, fundamental metallic sound
	Sheen/Sparkle	*4k - 12k*	For ride definition, crash brilliance
	Air/High Harmonics	*10k - 20k+*	Openness and shimmer
Congas			
	Body/Resonance	*150 - 300*	*Thump* or *ring* of the drum

Instrument Part	Aspect	Typical Frequency Range (Hz)	Notes & Characteristics
	Slap/Attack	*2k - 5k*	Sharp, cutting sound of the hand slap
Bongos			
	Body/Warmth	*120 - 250*	Fundamental tone
	Attack/Definition	*3k - 5k*	*Pop* or *crack* sound
Djembes			
	Body/Depth	*120 - 250*	Deep bass tones
	Attack/Clarity	*1k - 3k*	Definition of the strike
	High-End/Brightness	*8k - 10k*	Upper harmonics and crispness
Tabla			
	Fundamental (Dayan)	*200+*	Higher pitched drum, strong harmonics
	Fundamental (Bayan)	*50 - 150*	Lower, resonant bass drum
	Mid-range Harmonics	*500 - 2k*	Crucial for unique timbre of different strokes
	Attack/Definition	*2k - 5k*	Clarity of the strike
Shakers/Maracas			
	Main Sound/Sizzle	*4k - 15k*	Dominant *shhh* or *rattle* sound
	Overall Range	*200 - 18k+*	Varies by material and size; lower end for body
Tambourine			
	Jingle Sound/Sizzle	*5k - 15k*	Bright, metallic shimmer
	Overall Resonances	*2k - 8k*	Interaction of jingles and drumhead
Cowbell			
	Main Tone/Clang	*500 - 7k*	Distinct mid-range to high-mid presence
	Harmonics/Ring	*7k - 15k+*	Upper overtones
Wood Block			

Instrument Part	Aspect	Typical Frequency Range (Hz)	Notes & Characteristics
	Main Tone	*800 - 5k*	Sharp, resonant *clack* or *knock*
	Attack/Definition	*High Freq*	Strong transients
Claves			
	Main Tone/Click	*1k - 5k*	Very percussive, sharp *click*
Guiro			
	Scratch/Main Sound	*1k - 8k*	Characteristic *scrape* sound
Triangles			
	Fundamental/Ring	*1k - 8k*	Clear pitched ring
	Harmonics/Sustain	*10k - 20k+*	Shimmering, sustained high-frequency content
Chimes/Bells			
	Ring/Body	*1k - 5k*	Initial strike and primary resonance
	Sparkle/Air	*5k - 15k+*	Shimmering, sustained high-frequency content